SHREE RAJESH LONDHE

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PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer specializing in Generative AI, Multi-Agent workflows, and advanced RAG architectures. Proven track record in fine-tuning open-source LLMs (Mistral), constructing high-performance hybrid retrieval pipelines, and deploying scalable, state-managed agentic workflows using LangGraph and FastAPI. Focused on bridging the gap between deep learning research and production-grade, containerized MLOps deployments.

TECHNICAL SKILLS

Programming & Core Tools: Python, C++, C, SQL, Git, GitHub, Pandas, NumPy, Pydantic
Machine & Deep Learning: PyTorch, TensorFlow, Scikit-learn, HF Transformers, PEFT, LoRA, CrossEncoder, bitsandbytes
Generative AI & LLMs: LangGraph (StateGraph), LangChain, LlamaIndex, CrewAI, Multi-Agent Orchestration, OpenAI API, Gemini 2.0, QLoRA
Storage & Retrieval: Qdrant Cloud, FAISS, ChromaDB, Pinecone, PostgreSQL, BM25, Sparse/Dense Hybrid Retrieval
MLOps & Deployment: FastAPI, Docker, MLflow, Weights & Biases, Gradio, Streamlit, Railway, Vercel, fpdf2

PROFESSIONAL EXPERIENCE

Qriocity Ventures | *ML Developer (Remote)* — **Jan 2026 – May 2026**

- Developed and optimized end-to-end ML pipelines by preprocessing large-scale datasets, reducing high-frequency data noise, and engineering robust features to maximize downstream model training stability and performance metrics.
- Integrated Generative AI frameworks and open LLM APIs into core client applications, utilizing targeted context engineering and vector database indexing structures to dramatically scale search and information retrieval capabilities.
- Collaborated on deploying containerized production models utilizing Docker and FastAPI microservices, establishing robust API endpoints to ensure low-latency inference cycles and seamless integration with front-end micro-architectures.

TECHNICAL PROJECTS

Multi-Agent Research Assistant | *LangGraph, Gemini 2.0, Tavily, ArXiv* — github.com/Shree-2004/Multi-Agent-Research-Assistant

- **Architected a production-grade multi-agent system** using LangGraph **StateGraph**, mapping complex logic across 4 specialized agents (Researcher, Analyst, Writer, Critic) using unified state memory with **TypedDict** and custom state reducers.
- **Engineered a robust reflection loop** where a Critic agent executes a 5-point verification checklist, programmatically routing conditional execution paths back to the Analyst for up to 2 automated self-correction cycles to minimize factual gaps.
- **Built a custom latency and quality benchmark suite** using Pandas to record run-time execution, citation frequencies, and outputs, persisting detailed MLOps run-logs to track system degradation and performance drift.

SEBI/BSE Financial Filing RAG Analyzer | *Qdrant, FastAPI, BM25, PyTorch* — github.com/Shree-2004/SEBI-BSE-Financial-Filing-RAG

- **Engineered a hybrid retrieval mechanism** combining dense vector search (**BAAI/bge-base-en-v1.5**) and sparse BM25 keyword matching, optimizing precision over both semantic concepts and exact tabular financial figures.
- **Designed a structure-aware ingestion pipeline** with pdfplumber and Tesseract OCR, ensuring dense tables and complex text layout hierarchies remain structurally sound during high-fidelity recursive chunking.
- **Optimized generation accuracy** by applying a CrossEncoder re-ranking step and developing a verification layer that cross-references LLM outputs against retrieved text slices to enforce per-page citations.

Domain Fine-Tuned Finance LLM | *Mistral-7B, QLoRA, MLflow, HF* — github.com/Shree-2004/Finance-llm

- **Fine-tuned Mistral-7B-Instruct** on custom localized Indian personal finance data utilizing QLoRA (4-bit NormalFloat via bitsandbytes), adapting the LLM's reasoning engine while reducing memory footprint by 4×.
- **Conducted a systematic LoRA rank ablation study** (**r=8, 16, 32**) tracked inside MLflow to optimize gradient scaling parameters, and built an automated evaluation script yielding a **+88.6% improvement** in target keyword coverage.
- **Successfully merged and deployed** low-rank adapters back into the 16-bit base model, pushing the final functional weights directly to HuggingFace Hub and building an interactive Gradio web app.

EDUCATION

Dr. Babasaheb Ambedkar Technological University (DBATU) | B.Tech — IT (AI & ML Specialization) — **Expected Aug 2026**

CERTIFICATIONS & LANGUAGES

Certifications: Data Science with Generative AI (PWSkills) | Python for Data Science (Kaggle) | AI / ML (Apna College)
Languages: English (Fluent), Hindi (Fluent), Marathi (Native)